

3D-PLANES



SYNOPSIS

Equation of a Plane:

- Every first degree equation in x, y, z always represents a plane.
- Plane surface is a surface in which line joining every two points P and Q on it lies entirely in the surface.
- The general form of equation of plane is $ax + by + cz + d = 0$, a, b, c are not all zero i.e., $a^2 + b^2 + c^2 \neq 0$

Equation of Planes with Different Conditions:

- i) The equation of the plane passing through the point (x_1, y_1, z_1) and having d.r's of normal as (a, b, c) is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$ or $ax + by + cz = ax_1 + by_1 + cz_1$
- ii) The equation of the plane passing through a point (x_1, y_1, z_1) and parallel to the plane $ax + by + cz + d = 0$ is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$
 $\Rightarrow ax + by + cz = ax_1 + by_1 + cz_1$

W.E-1 : The equation of the plane parallel to the plane $2x + 3y + 4z + 5 = 0$ and passing through the point $(1, 1, 1)$ is

Sol : The plane is

$$\begin{aligned} a(x - x_1) + b(y - y_1) + c(z - z_1) &= 0 \\ 2(x - 1) + 3(y - 1) + 4(z - 1) &= 0 \\ \Rightarrow 2x + 3y + 4z - 9 &= 0 \end{aligned}$$

Equation of plane which is Parallel to lines:

- i) The equation of the plane passing through the point (x_1, y_1, z_1) and parallel to lines whose d.r's are (a_1, b_1, c_1) and (a_2, b_2, c_2) is

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

- ii) The equation of the plane passing through the points (x_1, y_1, z_1) , (x_2, y_2, z_2) and parallel to the line whose d.r's are (a, b, c) is

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a & b & c \end{vmatrix} = 0$$

- iii) The equation of the plane passing through three non collinear points

(x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) is

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

- iv) If (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) and (x_4, y_4, z_4) are coplanar, then

$$\begin{vmatrix} x_4 - x_1 & y_4 - y_1 & z_4 - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

General equation of a plane with different conditions:

- i) The equation of a plane with d.r's of normal as (a, b, c) is $ax + by + cz + d = 0$.
- ii) If a) $a = 0, b \neq 0, c \neq 0$ Then equation $by + cz + d = 0$ represents a plane which is parallel to x-axis and $\perp^{\text{er}}$ to YZ - plane.
 b) $b = 0, a \neq 0, c \neq 0$ then equation $ax + cz + d = 0$ represents a plane which is parallel to y-axis and $\perp^{\text{er}}$ to xz -plane.
 c) $a \neq 0, b \neq 0, c = 0$ then equation $ax + by + d = 0$ represents a plane which is parallel to z-axis and $\perp^{\text{er}}$ to XY -plane.
- iii) The equation of the plane passing through (x_1, y_1, z_1) and parallel to
 a) yz- plane and $\perp^{\text{er}}$ to X-axis is $x = x_1$
 b) xy-plane and $\perp^{\text{er}}$ to Z-axis is $z = z_1$
 c) zx-plane and $\perp^{\text{er}}$ to Y-axis is $y = y_1$

iv) Equation of plane parallel to the plane

$ax + by + cz + d_1 = 0$ is of the form

$$ax + by + cz + d_2 = 0$$

v) Distance between the above two parallel planes

$$\text{is } \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$$

vi) Equation of plane parallel to $\vec{r} \cdot \vec{n} = d_1$ is $\vec{r} \cdot \vec{n} = d_2$ (vector form)

vii) The equation of the plane, mid way between the parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is

$$ax + by + cz + \left(\frac{d_1 + d_2}{2}\right) = 0$$

viii) The equation of the plane which bisects the line joining $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ and perpendicular to AB is

$$\frac{(x_1 - x_2)x + (y_1 - y_2)y + (z_1 - z_2)z}{2} = \frac{(x_1^2 + y_1^2 + z_1^2) - (x_2^2 + y_2^2 + z_2^2)}{2}$$

ix) The reflection of $a^1x + b^1y + c^1z + d^1 = 0$ in the plane $ax + by + cz + d = 0$ is given by

$$2(aa^1 + bb^1 + cc^1)(ax + by + cz + d) = (a^2 + b^2 + c^2)(a^1x + b^1y + c^1z + d^1)$$

W.E-2 : Distance between parallel planes

$2x - 2y + z + 3 = 0$ and $4x - 4y + 2z + 5 = 0$ is

Sol : The given planes are $2x - 2y + z + 3 = 0$,

$$2x - 2y + z + \frac{5}{2} = 0$$

Here $a = 2, b = -2, c = 1, d_1 = 3, d_2 = \frac{5}{2}$

$$\text{Distance} = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}} = \frac{\left|3 - \frac{5}{2}\right|}{\sqrt{4 + 4 + 1}} = \frac{1}{6}$$

W.E-3 : The equation of the parallel plane lying midway between the parallel planes

$2x - 3y + 6z - 7 = 0$ and $2x - 3y + 6z + 7 = 0$ is

Sol : The required plane is $ax + by + cz + \frac{d_1 + d_2}{2} = 0$

$$2x - 3y + 6z + \frac{(-7 + 7)}{2} = 0$$

$$2x - 3y + 6z = 0$$

W.E-4 : The reflection of the plane in the plane

$x - y + z - 3 = 0$ is

Sol : The given planes are,

$$2x - 3y + 4z - 3 = 0 \quad (a^1x + b^1y + c^1z + d^1 = 0)$$

$$x - y + z - 3 = 0 \quad (ax + by + cz + d = 0)$$

$\therefore$ Equation of the required plane be obtained using the fact reflection of

$$a^1x + b^1y + c^1z + d^1 = 0 \text{ in the plane}$$

$ax + by + cz + d = 0$ is given by

$$2(aa^1 + bb^1 + cc^1)(ax + by + cz + d) = (a^2 + b^2 + c^2)(a^1x + b^1y + c^1z + d^1)$$

$$\therefore \text{The reflection is } 4x - 3y + 2z - 15 = 0$$

Foot and image:

➤ i) The foot of the perpendicular of the point $P(x_1, y_1, z_1)$ on the plane

$ax + by + cz + d = 0$ is $Q(h, k, l)$ then

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{l - z_1}{c} = \frac{-(ax_1 + by_1 + cz_1 + d)}{a^2 + b^2 + c^2}$$

ii) If $Q(h, k, l)$ is the image of the point

$P(x_1, y_1, z_1)$ w.r.to the plane $ax + by + cz + d = 0$

then

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{l - z_1}{c} = \frac{-2(ax_1 + by_1 + cz_1 + d)}{a^2 + b^2 + c^2}$$

iii) If 'd' is the distance from the origin and (l, m, n)

are the dc's of the normal to the plane through the

origin, then the foot of the perpendicular is

$$(ld, md, nd)$$

W.E-5 : The foot of the perpendicular from the point $P(1, 3, 4)$ to the plane $2x - y + z + 3 = 0$ is

Sol : The given plane is $2x - y + z + 3 = 0$

Here $a = 2, b = -1, c = 1, d = 3$,

$$(x_1, y_1, z_1) = (1, 3, 4)$$

$$\therefore \frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{l - z_1}{c} = \frac{-(ax_1 + by_1 + cz_1 + d)}{a^2 + b^2 + c^2}$$

$$\therefore (h, k, l) = (-1, 4, 3)$$

W.E-6 : If the image of the point $(-1, 3, 4)$ in the plane $x - 2y = 0$ is (x_1, y_1, z_1) then $z_1 =$

Sol : Given that $(h, k, l) = (x_1, y_1, z_1)$

$$(x_1, y_1, z_1) = (-1, 3, 4)$$

$$\therefore \frac{x_1 + 1}{1} = \frac{y_1 - 3}{-2} = \frac{z_1 - 4}{0} = - \left\{ \frac{(-1) - (2)(3)}{1^2 + (-2)^2 + 0^2} \right\}$$

$$\Rightarrow z_1 - 4 = 0, \Rightarrow z_1 = 4$$

Ratio formula:

- i) The ratio in which the plane $ax + by + cz + d = 0$ divides the line segment joining

(x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$-(ax_1 + by_1 + cz_1 + d) : (ax_2 + by_2 + cz_2 + d)$$

- ii) Position of the points w.r.to the plane

- a) If $\frac{ax_1 + by_1 + cz_1 + d}{ax_2 + by_2 + cz_2 + d} > 0$ then the points

$P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ lie on same side of the plane $ax + by + cz + d = 0$

- b) If $\frac{ax_1 + by_1 + cz_1 + d}{ax_2 + by_2 + cz_2 + d} < 0$ then the points

$P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ lie on opposite sides of the plane $ax + by + cz + d = 0$

W.E-7 : If the plane $2x - 3y + 5z - 2 = 0$ divides the line segment joining $(1, 2, 3)$ and $(2, 1, k)$ in the ratio $9 : 11$ then $k =$

Sol :
$$\frac{-(ax_1 + by_1 + cz_1 + d)}{ax_2 + by_2 + cz_2 + d} = \frac{9}{11}$$

$$\frac{(2 \times 1 - 3 \times 2 + 5 \times 3 - 2)}{(2 \times 2 - 3 \times 1 + 5 \times k - 2)} = \frac{9}{11}, \therefore k = -2$$

Normal form of a plane:

- i) If (l, m, n) are the direction cosines of normal to plane π and p is the $\perp^{\text{er}}$ distance from origin to the plane then the equation of plane is $lx + my + nz = p$

ii) The normal form of the plane representing by the equation $ax + by + cz + d = 0$ is

- a) If $d < 0$

$$\frac{a}{\sqrt{a^2 + b^2 + c^2}}x + \frac{b}{\sqrt{a^2 + b^2 + c^2}}y + \frac{c}{\sqrt{a^2 + b^2 + c^2}}z = \frac{-d}{\sqrt{a^2 + b^2 + c^2}}$$

- b) If $d > 0$

$$\frac{-a}{\sqrt{a^2 + b^2 + c^2}}x + \frac{-b}{\sqrt{a^2 + b^2 + c^2}}y + \frac{-c}{\sqrt{a^2 + b^2 + c^2}}z = \frac{d}{\sqrt{a^2 + b^2 + c^2}}$$

W.E-8 : If the equation of the plane $2x - 3y + 6z = 7$ in the normal form is

$$lx + my + nz = p \text{ then } l + p =$$

Sol : Equation of the plane in the normal form is

$$\frac{2x}{\sqrt{4+9+36}} + \frac{-3y}{\sqrt{4+9+36}} + \frac{6z}{\sqrt{4+9+36}} = \frac{7}{\sqrt{4+9+36}}$$

$$\frac{2}{7}x - \frac{3}{7}y + \frac{6}{7}z = 1, (lx + my + nz = p)$$

$$\Rightarrow l + p = \frac{2}{7} + 1 = \frac{9}{7}$$

Perpendicular distance from point to the plane:

- i) The perpendicular distance from (x_1, y_1, z_1) to

the plane $ax + by + cz + d = 0$ is $\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$

- ii) The perpendicular distance of the plane

$ax + by + cz + d = 0$ from the origin is $\frac{|d|}{\sqrt{a^2 + b^2 + c^2}}$

W.E-9 : If the perpendicular distance from $(1, 2, 4)$ to the plane $2x + 2y - z + k = 0$ is 3 then $k - 4 =$

Sol :
$$\left| \frac{2 + 4 - 4 + k}{\sqrt{4 + 4 + 1}} \right| = 3 \Rightarrow K = 7, k - 4 = 7 - 4 = 3$$

Intercept form of a plane:

- i) If a plane cuts X-axis at $A(a, 0, 0)$, Y-axis at $B(0, b, 0)$ and Z-axis at $C(0, 0, c)$ then a, b, c are called X-intercept, Y-intercept, Z-intercept of the plane.

- ii) The equation of the plane in intercept form is

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

- iii) If $ax + by + cz + d = 0$ is a plane if

$$a \neq 0, b \neq 0, c \neq 0 \text{ then X-intercept} = -\frac{d}{a}$$

$$\text{Y-intercept} = -\frac{d}{b}, \quad \text{Z-intercept} = -\frac{d}{c}$$

- iv) The equation of the plane whose intercepts are K times the intercepts made by the plane

$ax + by + cz + d = 0$ on corresponding axes is
 $ax + by + cz + kd = 0$.

Areas:

➤ i) Area of the triangle formed by the plane

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \text{ with}$$

a) X-axis, Y-axis is $\frac{1}{2}|ab|$ Sq. units

b) Y-axis, Z-axis is $\frac{1}{2}|bc|$ Sq. units

c) Z-axis, X-axis is $\frac{1}{2}|ca|$ Sq. units

ii) If the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ meets the co-ordinate axes in the points A, B, C. then the area of the triangle ABC is

$$\frac{1}{2}\sqrt{(ab)^2 + (bc)^2 + (ca)^2}.$$

W.E-10 : The area of the triangle formed by the plane $2x + 3y + 6z + 9 = 0$ with Y-axis, Z-axis is (in Sq.units)

Sol : The plane is $\frac{x}{-9/2} + \frac{y}{-3} + \frac{z}{-3/2} = 1$

Here $a = \frac{-9}{2}, b = -3, c = \frac{-3}{2}$

The area of the triangle = $\frac{1}{2}|bc|$
 $= \frac{1}{2} \left| (-3) \left(\frac{-3}{2} \right) \right| = \frac{9}{4}$ Sq.units

W.E-11 : The plane $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$ cuts the axes in A, B, C then the area of the $\triangle ABC$ is (sq)

Sol : $a = 2, b = 3, c = 4$

$\therefore$ Area of the $\triangle ABC =$

$$\frac{1}{2}\sqrt{(ab)^2 + (bc)^2 + (ca)^2}, = \sqrt{61}.$$

Angle between Two Planes:

➤ i) The angle between two planes is equal to the angle between the perpendiculars from the origin to the planes.

ii) If ' θ ' is the angle between the planes
 $a_1x + b_1y + c_1z + d_1 = 0$ and
 $a_2x + b_2y + c_2z + d_2 = 0$ then

$$\cos \theta = \pm \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

iii) If the above two planes are parallel then

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

iv) If the above two planes are perpendicular then
 $a_1a_2 + b_1b_2 + c_1c_2 = 0$

v) Angle between the line with d.c's (l_1, m_1, n_1) and the plane whose normal with d.c's (l_2, m_2, n_2) is θ then $\cos(90 - \theta) = |l_1l_2 + m_1m_2 + n_1n_2|$

vi) If θ is angle between a line L and a plane π then the angle between L and normal to the plane π is $90 \pm \theta$.

W.E-12 : If the angle between $x + 2y + 4z + 7 = 0$

and $x - 3y + 2z + 6 = 0$ is $\cos^{-1}(\sqrt{k})$, then $k =$

Sol : $\cos \theta = \left| \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} \right|$

Where $a_1 = 1; b_1 = 2; c_1 = 4$ and $a_2 = 1; b_2 = -3; c_2 = 2$

$$\therefore \cos \theta = \frac{|1 \times 1 + (2)(-3) + (4)(2)|}{\sqrt{1^2 + 2^2 + 4^2} \sqrt{1^2 + (-3)^2 + (2)^2}}$$

$$= \frac{3}{\sqrt{21}\sqrt{14}} = \sqrt{\frac{9}{21 \times 14}} = \sqrt{\frac{3}{98}}, \therefore k = \frac{3}{98}$$

W.E-13 : If the planes $2x - y - 3z - 7 = 0$ and

$4x - 2y + 5kz + 9 = 0$ are parallel then $5k^2 + 6 =$

Sol : $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$$\frac{2}{4} = \frac{-1}{-2} = \frac{-3}{5k} \Rightarrow k = \frac{-6}{5} \Rightarrow 5k^2 + 6 = \frac{66}{5}$$

W.E-14 : If the planes $x + 2y + kz = 0$ and

$2x + y - 2z = 0$ are at right angles, then the value of k is

Sol : $a_1a_2 + b_1b_2 + c_1c_2 = 0$ $1(2) + 2(1) + k(-2) = 0$
 $\Rightarrow 4 - 2k = 0, \Rightarrow k = 2$

Equations of planes bisecting the angles between given planes:

- i) Equations of two planes bisecting the angles between the planes $a_1x + b_1y + c_1z + d_1 = 0$ and

$$a_2x + b_2y + c_2z + d_2 = 0 \text{ are}$$

$$\frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \pm \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

ii) If $d_1, d_2 > 0$

Condition **Acute** **Obtuse**

$$a_1a_2 + b_1b_2 + c_1c_2 > 0 \quad - \quad +$$

$$a_1a_2 + b_1b_2 + c_1c_2 < 0 \quad + \quad -$$

iii) a) The Bisector planes are perpendicular to each other

b) Positive sign bisector is the bisector containing the origin.

The projection of line segment on a line (plane):

- Let P, Q be two points and L be a line (Π plane). If M, N are feet of perpendiculars from P, Q to the line L (to the plane Π) respectively then MN is called projection of PQ on the line L (the plane Π). The length of projection of PQ is always non-negative.

Some standard results:

- i) If $P_1 = a_1x + b_1y + c_1z + d_1 = 0$ and $P_2 = a_2x + b_2y + c_2z + d_2 = 0$ are two intersecting planes then the plane passing through their line of intersection is $P_1 + kP_2 = 0$ where k is any constant.
- ii) The equation of plane which bisects the join of the points (x_1, y_1, z_1) and (x_2, y_2, z_2) at right

$$\text{angles is } \sum (x_1 - x_2) \left\{ x - \frac{1}{2}(x_1 + x_2) \right\} = 0$$

iii) If a plane meets the coordinate axes in A, B, C such that the centroid of the triangle ABC is the point (p, q, r) then the equation of the plane is

$$\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 3$$

iv) Two systems of rectangular axes have the same origin. If a plane cuts them at distance a, b, c and a_1, b_1, c_1 respectively from the origin, then

$$a^{-2} + b^{-2} + c^{-2} = a_1^{-2} + b_1^{-2} + c_1^{-2}$$

v) A variable plane is at a constant distance 'p' from the origin and meets the axes in A, B and C. The locus of the centroid of the triangle ABC is

$$x^{-2} + y^{-2} + z^{-2} = 9p^{-2}$$

vi) A variable plane is at a constant distance 'p' from the origin and meets the axes in A, B and C. The locus of the centroid of the tetrahedron OABC is

$$x^{-2} + y^{-2} + z^{-2} = 16p^{-2}$$

vii) A variable plane passes through a fixed point (α, β, γ) and meets the coordinate axes in A, B, C. Then the locus of the point of intersection of the planes through A, B, C parallel to the coordinate

$$\text{planes is } \frac{\alpha}{x} + \frac{\beta}{y} + \frac{\gamma}{z} = 1$$

viii) A variable plane at a constant distance 'p' from the origin meets the axes in A, B and C. Through A, B, C planes are drawn parallel to the coordinate planes. Then the locus of their point of intersection is

$$x^{-2} + y^{-2} + z^{-2} = p^{-2}$$

ix) A point P moves on the fixed plane

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. The plane through P, perpendicular to OP meets the coordinate axes in A, B and C. Then the locus of the point of intersection of the planes through A, B, C parallel to the coordinate

$$\text{planes is } \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{ax} + \frac{1}{by} + \frac{1}{cz}.$$

x) The planes $x = \pm a, y = \pm b$ and $z = \pm c$ form a rectangular parallelepiped.

xi) A parallelepiped is formed by the planes drawn through the points (x_1, y_1, z_1) and (x_2, y_2, z_2) parallel to the co-ordinate planes. The length of a diagonal of the parallelepiped is

$$= \sqrt{a^2 + b^2 + c^2}$$

$$\text{Here } a = x_2 - x_1, b = y_2 - y_1, c = z_2 - z_1$$

**LEVEL-I(C.W.)****CARTESIAN EQUATION OF A PLANE**

1. The plane which passes through the point $(-1, 0, -6)$ and perpendicular to the line whose d.r's are $(6, 20, -1)$ also passes through the point

1) $(1, 1, -26)$

2) $(0, 0, 0)$

3) $(2, 1, -32)$

4) $(1, 1, 1)$

2. Equation of the plane through the mid-point of the join of A(4,5,-10) and B(-1,2,1) and perpendicular to AB is

1) $5x + 3y - 11z + \frac{135}{2} = 0$

2) $5x + 3y - 11z = \frac{135}{2}$

3) $5x + 3y + 11z = 135$

4) $5x + 3y - 11z + \frac{185}{2} = 0$

3. In the space the equation $by + cz + d = 0$ represents a plane perpendicular to the plane
1) YOZ 2) ZOY 3) XOY 4) Z = k

4. The equation of the plane passing through a point with position vector $\hat{i} + 2\hat{j} + 3\hat{k}$ and parallel to the plane $\vec{r} \cdot (3\hat{i} + 4\hat{j} + 5\hat{k}) = 0$ is

1) $3x + 4y - 5z + 26 = 0$

2) $3x + 4y + 5z - 26 = 0$

3) $3x - 4y + 5z - 26 = 0$

4) $3x + 4y - 5z - 26 = 0$

5. Distance between two parallel planes $2x + y + 2z = 8$ and $4x + 2y + 4z + 5 = 0$ is
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1) $\frac{3}{2}$ 2) $\frac{9}{2}$ 3) $\frac{7}{2}$ 4) $\frac{5}{2}$

FOOT AND IMAGE

6. If the foot of perpendicular from (0, 0, 0) to a plane is (1, 2, 2) then the equation of the plane is

1) $-x + 2y + 8z - 9 = 0$ 2) $x + 2y + 2z - 9 = 0$

3) $x + y + z - 5 = 0$ 4) $x + 2y = 3z + 1 = 0$

7. The foot of the perpendicular from (7, 14, 5) to $2x + 4y - z = 2$ is

1) (-1, 1, 0) 2) (1, 2, 8) 3) (2, -1, -2) 4) (1, 2, 3)

8. The image of the point (-1, 3, 4) in the plane $x - 2y = 0$ is

1) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$ 2) $\left(\frac{9}{5}, -\frac{13}{5}, 4\right)$

3) $\left(-\frac{17}{3}, -\frac{19}{3}, 4\right)$ 4) (15, 11, 4)

RATIO FORMULA

9. If the points (1, -1, 1) and (-2, 0, 5) with respect to the plane $2x + 3y - z + 7 = 0$ lie on
1) opposite sides 2) same side
3) on the plane 4) in side the plane

10. The ratio in which the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + 3\hat{k}) = 17$ divides the line joining the points $-2\hat{i} + 4\hat{j} + 7\hat{k}$ and $3\hat{i} - 5\hat{j} + 8\hat{k}$ is
1) 1 : 5 2) 1 : 10 3) 3 : 5 4) 3 : 10

NORMAL FORM OF A PLANE

11. For the plane $\Pi \equiv 2x + 3y + 5z + 10 = 0$, the point (2, 3, -5) lie in the
1) Opposite to the origin side
2) Origin side 3) Plane 4) can not say

12. The normal form of $2x - 2y + z = 5$ is

1) $12x - 4y + 3z = 39$ 2) $-\frac{6}{7}x + \frac{2}{7}y + \frac{3}{7}z = 1$

3) $\frac{12}{13}x - \frac{4}{13}y + \frac{3}{13}z = 3$ 4) $\frac{2}{3}x - \frac{2}{3}y + \frac{1}{3}z = \frac{5}{3}$

PERPENDICULAR DISTANCE

13. A plane passes through (2, 3, -1) and is perpendicular to the line having dr's (3, -4, 7). The perpendicular distance from the origin to this plane is (EAM-2011)

1) $\frac{3}{\sqrt{74}}$ 2) $\frac{5}{\sqrt{74}}$ 3) $\frac{6}{\sqrt{74}}$ 4) $\frac{13}{\sqrt{74}}$

14. An equation of a plane parallel to the plane $x - 2y + 2z - 5 = 0$ and at a unit distance from the origin is (AIE-2012)

1) $x - 2y + 2z - 7 = 0$ 2) $x - 2y + 2z + 5 = 0$

3) $x - 2y + 2z - 3 = 0$ 4) $x - 2y + 2z + 1 = 0$

INTERCEPT FORM OF A PLANE

15. The equation of a plane which cuts equal intercepts of unit length on the axes is

1) $x + y + z = 0$ 2) $x + y + z = 1$

3) $2x + y - z = 1$ 4) $\frac{x}{a} + \frac{y}{a} + \frac{z}{a} = 1$

16. If the plane $7x + 11y + 13z = 3003$ meets the coordinate axes in A, B, C then the centroid of the $\triangle ABC$ is

1) (143, 91, 77) 2) (143, 77, 91)

3) (91, 143, 77) 4) (143, 66, 91)

17. If the areas of triangles formed by a plane with the positive X, Y, Z, X axes respectively are 12, 9, 6 sq. unit respectively then the equation of the plane is

1) $\frac{x}{4} + \frac{y}{6} + \frac{z}{3} = 1$ 2) $\frac{x}{6} + \frac{y}{3} + \frac{z}{4} = 1$
 3) $\frac{x}{3} + \frac{y}{4} + \frac{z}{6} = 1$ 4) $\frac{x}{3} + \frac{y}{6} + \frac{z}{4} = 1$

ANGLE BETWEEN TWO PLANES

18. If the planes $x + 2y + kz = 0$ and $2x + y - 2z + 3 = 0$ are at right angles, then the values of k is

1) $-\frac{1}{2}$ 2) $\frac{1}{2}$ 3) -2 4) 2

19. The angle between the planes $2x + y + z = 3$, $x - y + 2z = 5$ is

1) $\frac{\pi}{2}$ 2) $\frac{\pi}{6}$ 3) $\frac{3\pi}{4}$ 4) $\frac{\pi}{3}$

20. If the planes $2x + 3y - z + 5 = 0$, $x + 2y - kz + 7 = 0$ are perpendicular then $k =$
 1) 4 2) 6 3) 8 4) -8

LEVEL-I (C.W)-KEY

1) 2 2) 2 3) 1 4) 2 5) 3 6) 2
 7) 2 8) 2 9) 1 10) 4 11) 1 12) 4
 13) 4 14) 3 15) 2 16) 1 17) 1 18) 4
 19) 4 20) 4

LEVEL-I (C.W)-HINTS

- The plane equation is
 $6(x+1) + 20(y-0) - (z+6) = 0$
 $\Rightarrow 6x + 20y - z = 0$
- Mid point of AB = $\left(\frac{3}{2}, \frac{7}{2}, \frac{-9}{2}\right)$
 D.r's of AB are (5, 3, -11)
 Equation of plane is

$$\left[\bar{r} - \left(\frac{3}{2}\bar{i} + \frac{7}{2}\bar{j} - \frac{9}{2}\bar{k}\right)\right] \cdot (5\bar{i} + 3\bar{j} - 11\bar{k}) = 0$$

$$\bar{r} \cdot (5\bar{i} + 3\bar{j} - 11\bar{k}) = \frac{135}{2}$$
- Verification method
- Equation of the plane parallel to
 $\bar{r} \cdot (3\bar{i} + 4\bar{j} + 5\bar{k}) = 0$ is given by
 $(x\bar{i} + y\bar{j} + z\bar{k}) \cdot (3\bar{i} + 4\bar{j} + 5\bar{k}) + \lambda = 0$

It passes through (1, 2, 3)

$\therefore \lambda = -26$

5. Given planes are $4x + 2y + 4z - 16 = 0$,
 $4x + 2y + 4z + 5 = 0$

Distance between the parallel planes

$$= \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}} = \frac{7}{2}$$

6. D.r's of the perpendicular to the plane are (1, 2, 2)
 7. D.r's of the normal to $2x + 4y - z = 2$ is (2, 4, -1)

The point (1, 2, 8) lies on a plane and D.r's of a line joining (1, 2, 8) and (7, 14, 5) are
 $(6, 12, -3) = (2, 4, -1)$.,

$\therefore$ Required point = (1, 2, 8)

8. Given $(x_1, y_1, z_1) = (-1, 3, 4)$

$$\therefore \frac{h+1}{1} = \frac{k-3}{-2} = \frac{l-4}{0} = -\frac{2(-1-2(3))}{1+4+0}$$

$$\Rightarrow \frac{h+1}{1} = \frac{k-3}{-2} = \frac{l-4}{0} = -\frac{2(-1-2(3))}{1+4+0}$$

$$\Rightarrow (h, k, l) = \left(\frac{9}{5}, \frac{-13}{5}, 4\right)$$

9. $\Pi_{111} = 2(1) + 3(-1) - 1 + 7 > 0$

$$\Pi_{222} = 2(-2) + 3(0) - 5 + 7 < 0$$

$\therefore$ Points lies on opposite sides of the plane

10. Plane is $\bar{r} \cdot (\bar{i} - 2\bar{j} + 3\bar{k}) = 17$ --(1)

A point P dividing the join of

$-2\bar{i} + 4\bar{j} + 7\bar{k}$ and $3\bar{i} - 5\bar{j} + 8\bar{k}$ in the ratio

$$\lambda : 1 \text{ is } \left(\frac{3\lambda-2}{\lambda+1}\right)\bar{i} + \left(\frac{-5\lambda+4}{\lambda+1}\right)\bar{j} + \left(\frac{8\lambda+7}{\lambda+1}\right)\bar{k}$$

It lies on (1) then we get, $\lambda = \frac{3}{10} \Rightarrow \lambda : 1 = 3 : 10$

11. $\Pi_{111} = 4 + 9 - 25 + 10 = -2$, $\Pi_{222} = 10 \Rightarrow$ The point lie on the opposite to the origin side.

12. $\sqrt{a^2 + b^2 + c^2} = \sqrt{4 + 4 + 1} = 3$.

Normal form is $\frac{2}{3}x - \frac{2}{3}y + \frac{1}{3}z = \frac{5}{3}$

13. Equation of the plane is
 $3(x-2)-4(y-3)+7(z+1)=0$
 $\Rightarrow 3x-4y+7z+13=0$
 Perpendicular distance from origin
 $= \frac{13}{\sqrt{9+16+49}} = \frac{13}{\sqrt{74}}$
14. The required plane is $x-2y+2z+k=0$.
 The distance to the plane from origin = 1
 $\Rightarrow \frac{|k|}{\sqrt{1+4+4}} = 1 \Rightarrow k = \pm 3$
15. Verification.
16. The plane $7x+11y+13z=3003$ meets the coordinate axes in $A(429,0,0)$, $B(0,273,0)$, $C(0,0,231)$.
 Centroid of $\triangle ABC$ is $\left(\frac{429}{3}, \frac{273}{3}, \frac{231}{3}\right)$
 $= (143, 91, 77)$
17. If a, b, c are the intercepts of the required plane
 then $\frac{1}{2}ab=12, \frac{1}{2}bc=9, \frac{1}{2}ca=6$
 $\Rightarrow ab=24, bc=18, ca=12$
 $a^2 = \frac{abac}{bc} = \frac{24 \times 12}{18} = 16 \Rightarrow a=4$
 $ab=24, ac=12 \Rightarrow b=6, c=3$
 $\therefore$ The equation of the plane is $\frac{x}{4} + \frac{y}{6} + \frac{z}{3} = 1$
18. Use $a_1a_2+b_1b_2+c_1c_2=0$
19. $\cos \theta = \frac{|(2)(1)+(1)(-1)+(1)(2)|}{\sqrt{4+1+1}\sqrt{1+1+4}} = \frac{3}{6} = \frac{1}{2}$
 $\Rightarrow \theta = \frac{\pi}{3}$
20. Given planes are perpendicular $a_1a_2+b_1b_2+c_1c_2=0$
 $\Rightarrow (2)(1)+(3)(2)+(-1)(-k)=0$
 $\Rightarrow 2+6+k=0, \Rightarrow k=-8$



LEVEL-I(H.W.)

CARTESIAN EQUATION OF A PLANE

1. The product of the d.r.'s of a line perpendicular to the plane passing through the points $(4,0,0)$, $(0,2,0)$ and $(1,0,1)$ is
 1) 6 2) 2 3) 0 4) 1

2. A plane which passes through the point $(3, 2, 0)$ and the line $\frac{x-4}{1} = \frac{y-7}{5} = \frac{z-4}{4}$ is
 1) $x-y+z=1$ 2) $x+y+z=5$
 3) $x-2y-z=1$ 4) $2x-y+z=5$
3. The equation of the plane parallel to the plane $2x+3y+4z+5=0$ and passing through the point $(1,1,1)$ is
 1) $2x+3y+4z-9=0$ 2) $2x+3y+4z+9=0$
 3) $2x+3y+4z+7=0$ 4) $2x+3y+4z-7=0$
4. The equation of the plane passing through the point $(3, -6, 9)$ and perpendicular to the x-axis is
 1) $x+2=0$ 2) $y-3=0$ 3) $z-7=0$ 4) $x-3=0$
5. Distance between two parallel planes $7x+4y-4z+3=0$ and $14x+8y-8z-12=0$ is
 1) $\frac{15}{9}$ 2) 1 3) $\frac{9}{15}$ 4) $\frac{1}{2}$

FOOT AND IMAGE

6. If the foot of the perpendicular from $(0,0,0)$ to a plane is $(1,2,3)$, then the equation of the plane is (EAM-2012)
 1) $2x+y+3z=14$ 2) $x+2y+3z=14$
 3) $x+2y+3z+14=0$ 4) $x+2y-3z=14$
7. The image of the point $(3,2,1)$ in the plane $2x-y+3z=7$ is (EAM-2009)
 1) $(1,2,3)$ 2) $(2,3,1)$ 3) $(3,2,1)$ 4) $(2,1,3)$
8. If $(2,3,-1)$ is the foot of the perpendicular from $(4,2,1)$ to a plane, the equation of the plane is
 1) $2x-y-2z-3=0$ 2) $2x+y-2z-9=0$
 3) $2x+y+2z-5=0$ 4) $2x-y+2z+1=0$
9. If the points $(1,1,p)$ and $(-3,0,1)$ be equidistant from the plane $\vec{r} \cdot (3\vec{i} + 4\vec{j} + 12\vec{k}) + 13 = 0$ then the value of P =
 1) $-\frac{1}{3}$ 2) 6 3) 3 4) $\frac{1}{3}$

RATIO FORMULA

10. The ratio in which the line joining $(2, -4, 3)$ and $(-4, 5, -6)$ is divided by the plane $3x+2y+z-4=0$ is (EAM-2011)
 1) 2:1 2) 4:3 3) -1:4 4) 2:3

11. For the plane $\Pi \equiv 4x - 3y + 2z - 3 = 0$, the points $A(-2, 1, 2)$, $B = (3, 1, -2)$

- 1) Lie on the same side of $\Pi = 0$
 2) Lie on the opposite sides of $\Pi = 0$
 3) Lie on the normal to $\Pi = 0$ 4) Lie on $\Pi = 0$

NORMAL FORM OF A PLANE

12. The d.c's of the normal to the plane $2x - y + 2z + 5 = 0$ are

- 1) $(3, -2, 6)$ 2) $\left(\frac{2}{7}, \frac{3}{7}, \frac{-6}{7}\right)$
 3) $\left(\frac{3}{7}, \frac{-2}{7}, \frac{6}{7}\right)$ 4) $\left(\frac{2}{3}, \frac{-1}{3}, \frac{2}{3}\right)$

PERPENDICULAR DISTANCE

13. The perpendicular distance from the origin to the plane $x - 2y + 2z - 9 = 0$ is

- 1) 3 2) 6 3) 4 4) 2

14. The length of the perpendicular from the point $(7, 14, 5)$ to the plane $2x + 4y - z = 2$ is

- 1) $21\sqrt{3}$ 2) $3\sqrt{21}$ 3) $\sqrt{321}$ 4) $5\sqrt{2}$

INTERCEPT FORM OF A PLANE

15. 5, 7 are the intercepts of a plane on the Y-axis, Z-axis respectively, if the plane is parallel to the X-axis then the equation of that plane is

- 1) $5y + 7z = 35$ 2) $7y + 5z = 1$
 3) $\frac{y}{5} + \frac{z}{7} = 35$ 4) $7y + 5z = 35$

16. The intercepts of the plane $2x - 3y + 5z - 30 = 0$ are

- 1) 15, -10, 6 2) 5, 10, 6
 3) $\frac{1}{8}, \frac{-1}{6}, \frac{1}{4}$ 4) 3, -4, 6

17. The area of the triangle formed by

$\frac{x}{4} + \frac{y}{3} - \frac{z}{2} = 1$ with X-axis and Y-axis is

1) 2 2) 3 3) 6 4) 12

ANGLE BETWEEN TWO PLANES

18. If the planes $2x + 3y + 4z + 7 = 0$ and $4x + ky + 8z + 1 = 0$ are parallel then $k =$

- 1) 2 2) 4 3) 5 4) 6

19. The planes $2x - y + 4z = 5$ and $5x - (2.5)y + 10z = 6$ are

- 1) Perpendicular 2) parallel
 3) intersect Y-axis 4) passes through $(0, 0, 5/4)$

20. If $\lambda x + 4y + 5z = 7$, $4x + 4\lambda y + 10z - 14 = 0$ represent the same plane then the value of $\lambda =$

- 1) 1 2) 2 3) 0 4) 3

LEVEL-I (H.W)-KEY

- 1) 1 2) 1 3) 1 4) 4 5) 2 6) 2
 7) 3 8) 4 9) 1 10) 3 11) 2 12) 4
 13) 1 14) 2 15) 4 16) 1 17) 3 18) 4
 19) 2 20) 2

LEVEL-I (H.W)-HINTS

- Equation of the plane passing through the given points is $x + 2y + 3z - 4 = 0$,
D. r's of normal = $(1, 2, 3)$
- Verify options
- $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$
- Equation of the plane passing through $(3, -6, 9)$ and perpendicular to x-axis is $x = x_1$
 $\Rightarrow x - 3 = 0$
- Distance = $\frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}} = 1$
- Equation of the plane is $1(x - 1) + 2(y - 2) + 3(z - 3) = 0$
 $\Rightarrow x + 2y + 3z = 14$
- $(3, 2, 1)$ lies in the plane $2x - y + 3z = 7$
 $\Rightarrow (3, 2, 1)$ it self is the image in $2x - y + 3z = 7$
- The line joining the given points is normal to the plane
 $\therefore$ D.r's of normal are $(2, -1, 2)$ and $(2, 3, -1)$ lies in the plane
 $\therefore$ The equation of the plane is $2x - y + 2z = 4 - 3 - 2$
 $\Rightarrow 2x - y + 2z + 1 = 0$
- Distance from point $A(\bar{a})$ to plane $\bar{r} \cdot \bar{n} + k = 0$
is $\frac{|\bar{a} \cdot \bar{n} + k|}{|\bar{n}|}$
- Required ratio = $-(6 - 8 + 3 - 4) : (-12 + 10 - 6 - 4)$
 $= 3 : -12 = -1 : 4$
- $\Pi_{111} = -8 - 3 + 6 - 3 = -8$
 $\Pi_{222} = 12 - 3 - 4 - 3 = 2$
The points lie on the opposite side of $\Pi = 0$

$$12. Dc's = \pm \left(\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}} \right) \\ = \pm \left(\frac{2}{3}, \frac{-1}{3}, \frac{2}{3} \right)$$

$$13. \text{ Perpendicular distance } p = \frac{|-9|}{\sqrt{1+4+4}} = 3$$

$$14. \text{ Perpendicular distance } p = \frac{|14+56-5-2|}{\sqrt{4+16+1}} = \frac{63}{\sqrt{21}} = 3\sqrt{21}$$

$$15. \text{ Plane equation is } \frac{y}{5} + \frac{z}{7} = 1$$

$$16. \text{ The plane is } 2x - 3y + 5z = 30. \text{ Intercepts are } \frac{30}{2}, \frac{30}{-3}, \frac{30}{5} = 15, -10, 6$$

$$17. \text{ Area} = \frac{1}{2} |4 \times 3| = 6 \text{ sq. units}$$

$$18. \text{ Given planes are parallel} \\ \Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \\ \Rightarrow \frac{2}{4} = \frac{3}{k} \Rightarrow 2k = 12 \Rightarrow k = 6$$

$$19. \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$20. \text{ Given equations represent the same plane} \\ \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{d_1}{d_2} \Rightarrow \frac{\lambda}{4} = \frac{5}{10} \Rightarrow \lambda = 2$$



LEVEL-II(C.W.)

CARTESIAN EQUATION OF A PLANE

- If the equation of the plane passing through the points (1,2,3), (-1,2,0) and perpendicular to the ZX - plane is $ax + by + cz + d = 0$ ($a > 0$) then
 - $a = 0$ and $c = 0$
 - $a + d = 0$
 - $c + d - 5 = 0$
 - $a + c + d - 4 = 0$
- The equation of the plane through the line of intersection of planes $ax + by + cz + d = 0$, $a'x + b'y + c'z + d' = 0$ and parallel to the lines $y = 0$ and $z = 0$ is
 - $(ab' - a'b)x + (bc' - b'c)y + (ad' - a'd)z = 0$
 - $(ab' - a'b)y + (ac' - a'c)z + (ad' - a'd)x = 0$

$$3) (ab' - a'b)x + (bc' - b'c)z + (ad' - a'd)y = 0$$

$$4) (ab' - a'b)x - (bc' - b'c)y + (ad' - a'd)z = 0$$

- A plane π passes through the point (1,1,1). If b, c, a are the dr's of a normal to the plane, where a, b, c ($a < b < c$) are the prime factors of 2001, then the equation of the plane π is
 - $29x + 31y + 3z = 63$
 - $23x + 29y - 29z = 23$
 - $23x + 29y + 3z = 55$
 - $31x + 27y + 3z = 71$

- The dr's of a normal to the plane through (1,0,0), (0,1,0) which makes an angle of $\frac{\pi}{4}$ with the plane $x + y = 3$ are

$$1) 1, \sqrt{2}, 1 \quad 2) 1, 1, \sqrt{2} \quad 3) 1, 1, 2 \quad 4) \sqrt{2}, 1, 1$$

- Let $A(1,1,1), B(2,3,5)$ and $C(-1,0,2)$ be three points, then equation of a plane parallel to the plane ABC which is at a distance 2 units is
 - $2x - 3y + z + 2\sqrt{14} = 0$
 - $2x - 3y + z - \sqrt{14} = 0$
 - $2x - 3y + z + 2 = 0$
 - $2x - 3y + z - 2 = 0$

INTERCEPT FORM OF A PLANE

- A variable plane intersects the coordinate axes at A, B, C and is at a constant distance 'p' from O(0,0,0). Then the locus of the centroid of the tetrahedron OABC is
 - $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{p^2}$
 - $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{4}{p^2}$
 - $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{16}{p^2}$
 - $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = 16p^2$
- The equation to the plane through the line of intersection of $2x + y + 3z - 2 = 0$, $x - y + z + 4 = 0$ such that each plane is at a distance of 2 unit from the origin is
 - $x + y + 2z + 13 = 0, x + y + z - 3 = 0$
 - $2x + y - 2z + 3 = 0, x - 2y - 2z - 3 = 0$
 - $15x - 12y + 16z + 50 = 0, x + 2y + 2z - 6 = 0$
 - $x - y + 2z - 13 = 0, x + y - z - 3 = 0$
- The equation of the plane which is parallel to X-axis and making intercepts 3 and 8 on Y and Z-axes respectively is
 - $3y + 8z = 24$
 - $3y - 8z = 24$
 - $8y - 3z = 24$
 - $8y + 3z = 24$
- The sum of the intercepts of the plane which bisects the line segment joining (0,1,2) and (2,3,0) perpendicularly is
 - 2
 - 4
 - 6
 - 12

10. A plane meets the coordinate axes at A, B, C so that the centroid of the triangle ABC is (1, 2, 4). Then the equation of the plane is (EAM-2010)

1) $x + 2y + 4z = 12$ 2) $4x + 2y + z = 12$
3) $x + 2y + 4z = 3$ 4) $4x + 2y + z = 3$

11. The reflection of the plane $2x - 3y + 4z - 3 = 0$ in the plane $x - y + z - 3 = 0$ is the plane

1) $4x - 3y + 2z - 15 = 0$ 2) $x - 3y + 2z - 15 = 0$
3) $4x + 3y - 2z + 15 = 0$ 4) $4x + 3y + 2z + 15 = 0$

LEVEL-II (C.W)-KEY

1) 4 2) 2 3) 3 4) 2 5) 1 6) 3
7) 3 8) 4 9) 1 10) 2 11) 1

LEVEL-II (C.W)-HINTS

- The plane equation is $\frac{x}{l} + \frac{z}{m} = 1$.
 $\frac{1}{l} + \frac{3}{m} = 1; \frac{-1}{l} = 1 \Rightarrow l = -1; m = 3/2$
- Equation of the plane through the intersection of the planes $ax + by + cz + d = 0$ and $a^1x + b^1y + c^1z + d^1 = 0$
 $(a^1x + b^1y + c^1z + d^1) + \lambda(ax + by + cz + d) = 0$
which is parallel to $Y = 0 = Z$
parallel to X - axis
 $\Rightarrow (a^1 + a\lambda)^1 = 0 \Rightarrow a\lambda = -a^1 = \lambda = \frac{-a^1}{a}$
the equation of the plane is
 $(a^1b - ab^1)y + (a^1c - ac^1)z + a^1d - ad^1 = 0$
- $2001 = 3 \times 23 \times 29$ and
 $(3 + 23 + 29) = 55 \Rightarrow a = 3, b = 23, c = 29$
- Any plane through (1,0,0) is
 $A(x-1) + B(y-0) + C(z-0) = 0 \dots\dots(1)$
It contains (0,1,0) if $-A + B = 0 \dots\dots(2)$
Also (1) makes an angle of $\frac{\pi}{4}$ with the plane
 $x + y = 3$,
 $\therefore \cos \frac{\pi}{4} = \frac{|A+B|}{\sqrt{A^2+B^2+C^2} \sqrt{1^2+1^2}}$

$\Rightarrow 2AB = C^2 \dots\dots(3)$

From (2) and (3),

$C^2 = 2A^2 \Rightarrow C = \pm\sqrt{2}A$

Hence $A:B:C = A:A:\pm\sqrt{2}A$

$\therefore$ D.r's are $(1, 1, \pm\sqrt{2})$

5. $A(1, 1, 1), B(2, 3, 5), C(-1, 0, 2)$ d.r's of AB are (1, 2, 4)

D.r's of AC are $(-2, -1, 1)$.

D.r's of normal to plane ABC are (2, -3, 1) As a result, equation of the plane ABC is $2x - 3y + z = 0$ Let the equation of the required plane is $2x - 3y + z = k$, then

$\left| \frac{k}{\sqrt{4+9+1}} \right| = 2k = \pm 2\sqrt{14}$

Hence, equation of the required plane is $2x - 3y + z + 2\sqrt{14} = 0$

6. Intercepts of the plane = 4x, 4y, 4z where (x,y,z) is the centroid of the tetrahedron OABC.

Here, $\frac{1}{16x_1^2} + \frac{1}{16y_1^2} + \frac{1}{16z_1^2} = \frac{1}{p^2}$

7. Equation of the plane is
 $(2x + y + 3z - 2) + k(x - y + z + 4) = 0$

$p = \frac{|-2+4k|}{\sqrt{(2+k)^2 + (1-k)^2 + (3+k)^2}} = 2$

$\Rightarrow k = 13, -1$

$\therefore$ Planes are $(2x + y + 3z - 2) + 13$

$(x - y + z + 4) = 0, (2x + y + 3z - 2) - 1$

$(x - y + z + 4) = 0$

$\Rightarrow 15x - 12y + 16z + 50 = 0, x + 2y + 2z - 6 = 0$

8. Required plane equation is

$\frac{y}{3} + \frac{z}{8} = 1 \Rightarrow 8y + 3z = 24$

9. The plane equation is

$2x(x_2 - x_1) + 2y(y_2 - y_1) + 2z(z_2 - z_1)$

$+ x_1^2 + y_1^2 + z_1^2 - x_2^2 - y_2^2 - z_2^2 = 0$

where $(x_1, y_1, z_1) = (0, 1, 2)$ and $(x_2, y_2, z_2) = (2, 3, 0)$

10. Let the equation of the plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

$$\therefore A = (a, 0, 0), B(0, b, 0), C(0, 0, c)$$

$$\text{Centroid of } \triangle ABC = (1, 2, 4)$$

$$\Rightarrow \left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right) = (1, 2, 4) \Rightarrow a = 3, b = 6, c = 12$$

$$\text{The equation of the plane is } \frac{x}{3} + \frac{y}{6} + \frac{z}{12} = 1$$

$$\Rightarrow 4x + 2y + z = 12$$

11. Equation of the required plane be obtained using the reflection of $a^1x + b^1y + c^1z + d^1 = 0$ in the plane $ax + by + cz + d = 0$ is given by

$$2(aa^1 + bb^1 + cc^1)(ax + by + cz + d) \\ = (a^2 + b^2 + c^2)(a^1x + b^1y + c^1z + d^1)$$



LEVEL-II(H.W.)

CARTESIAN EQUATION OF A PLANE

- The d.r.'s of the line of intersection of the planes $x + y + z - 1 = 0$ and $2x + 3y + 4z - 7 = 0$ are
1) 1, 2, -3 2) 2, 1, -3
3) 4, 2, -6 4) 1, -2, 1
- The vertices of a tetrahedron are A(3,4,2) B(1,2,1) C(4,1,3) D(-1,-1,3). The height of A above the base BCD.
1) $\frac{27}{\sqrt{237}}$ 2) $\frac{23}{\sqrt{237}}$ 3) $\frac{20}{\sqrt{237}}$ 4) $\frac{27}{\sqrt{247}}$
- The equation of the plane which passes through the line of intersection of the planes $2x - y = 0$ and $3z - y = 0$ and is perpendicular to the plane $4x + 5y - 3z = 8$ is
1) $28x - 17y + 9z = 0$ 2) $28x + 17y + 9z = 0$
3) $2x + 17y - 9z = 0$ 4) $2x - y - z = 0$
- The dr's of a normal to the plane passing through (0,0,1), (0,1,2) and (1,2,3) are
1) (0,1,-1) 2) (1,0,-1)
3) (0,0,-1) 4) (1,0,0)

5. The plane $2x + 3y + kz - 7 = 0$ is parallel to the line whose direction ratios are (2, -3, 1) then $k =$

- 1) 5 2) 8 3) 1 4) 0

INTERCEPT FORM OF A PLANE

- A variable plane is at a constant distance 3p from the origin and meets the axes in A, B and C. The locus of the centroid of the triangle ABC is
1) $x^{-2} + y^{-2} + z^{-2} = p^{-2}$
2) $x^{-2} + y^{-2} + z^{-2} = 4p^{-2}$
3) $x^{-2} + y^{-2} + z^{-2} = 16p^{-2}$
4) $x^{-2} + y^{-2} + z^{-2} = 9p^{-2}$
- The equation of the plane through the line of intersection of the planes $x - 2y + 3z - 1 = 0$, $2x + y + z - 2 = 0$ and the point (1, 2, 3) is
1) $7x - 9y + 8z = 0$ 2) $7x + y + 8z = 0$
3) $x + 3y - 2z - 1 = 0$ 4) $x - 3y - 2z + 1 = 0$
- The equation of the plane which is parallel to Y-axis and making intercepts of lengths 3 and 4 on X-axis and Z-axis is
1) $2x + 2z = 20$ 2) $4x + 3z = 12$
3) $4x - 3z = 12$ 4) $6x + 13z = 15$
- If 5, 7, 6 are the sums of the X, Y intercepts; Y, Z intercepts, Z, X intercepts respectively of a plane then the perpendicular distance from the origin to that plane is
1) $\frac{144}{61}$ 2) $\frac{12}{\sqrt{61}}$ 3) $\frac{\sqrt{61}}{12}$ 4) $\frac{61}{144}$
- The equation of the plane through the line of intersection of the planes $2x + 3y + 4z - 7 = 0$, $x + y + z - 1 = 0$ and perpendicular to the plane $x - 5y + 3z - 2 = 0$ is
1) $7x - y - 6z - 17 = 0$ 2) $x - y - 6z - 27 = 0$
3) $x + 2y + 3z - 6 = 0$ 4) $x + y + 6z - 27 = 0$

11. The equations of bisectors of angles between YZ-plane and XZ-plane is

- 1) $x - z = 0, x + 2z = 0$ 2) $x - z + 2 = 0$
3) $x + z = 0, x - z = 0$ 4) $x + y = 0, x - y = 0$

LEVEL-II (H.W)-KEY

- 1) 4 2) 2 3) 1 4) 1 5) 1 6) 1
7) 3 8) 2 9) 2 10) 3 11) 4

LEVEL-II (H.W)-HINTS

1. Let (a, b, c) be the D.r's of the line then
 $a + b + c = 0, 2a + 3b + 4c = 0$ solving
 $a = 1, b = -2, c = 1$.

$$\therefore (a, b, c) = (1, -2, 1)$$

2. The equation of the plane containing BCD

$$\begin{vmatrix} x-1 & y-2 & z-1 \\ 3 & -1 & 2 \\ -2 & -3 & 2 \end{vmatrix} = 0$$

$$\Rightarrow 4x - 10y - 11z + 27 = 0$$

$$\text{distance from } A = \frac{23}{\sqrt{237}}$$

3. Any plane through the line is

$$2x - y + \lambda(3z - y) = 0 \quad (1)$$

$$\text{given } 4x + 5y - 3z - 8 = 0 \quad (2)$$

(1) and (2) are perpendicular we have

$$2(4) + 5(-(1 + \lambda)) + 3\lambda(-3) = 0 \Rightarrow \lambda = \frac{3}{14}$$

$$\Rightarrow 14(2x - y) + 3(3z - y) = 0$$

$$\Rightarrow 28x - 17y + 9z = 0$$

4. Let $A(0, 0, 1), B(0, 1, 2), C(1, 2, 3)$

$$\overrightarrow{AB} = (0, 1, 1), \overrightarrow{AC} = (1, 2, 2),$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 1 \\ 1 & 2 & 2 \end{vmatrix} =$$

$$\vec{i}(2-2) - \vec{j}(0-1) + \vec{k}(0-1) = \vec{j} - \vec{k}$$

$\therefore$ D.r's of normal to the plane are $(0, 1, -1)$

5. D.r's of the normal to the plane are $(2, 3, k)$. This is parallel to the line having D.r's is $(2, -3, 1)$ then
 $2(2) + 3(-3) + k(1) = 0 \Rightarrow 4 - 9 + k = 0 \Rightarrow k = 5$

6. If $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

distance from origin $= 3p$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{9p^2}$$

$$\Rightarrow \text{Then } a = 3x_1, b = 3y_1, c = 3z_1$$

$$\therefore \text{Locus is } \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{p^2}$$

7. Equation of the plane is

$$(x - 2y + 3z - 1) + k(2x + y + z - 2) = 0$$

It passes through $(1, 2, 3)$ then

$$5 + 5k = 0 \Rightarrow k = -1$$

$\therefore$ Plane is

$$(x - 2y + 3z - 1) - 1(2x + y + z - 2) = 0$$

$$\Rightarrow x + 3y - 2z - 1 = 0$$

8. Equation of plane parallel to Y-axis is of the both

$$\frac{x}{a} + \frac{z}{c} = 1 \Rightarrow \frac{x}{3} + \frac{z}{4} = 1 \Rightarrow 4x + 3z = 12$$

9. $a + b = 5, b + c = 7, c + a = 6$

$$\Rightarrow a + b + c = 9. \Rightarrow a = 2, b = 3, c = 4.$$

10. Equation of the plane is

$$(2x + 3y + 4z - 7) + k(x + y + z - 1) = 0$$

It is perpendicular to $x - 5y + 3z - 2 = 0$ then

$$\Rightarrow -k - 1 = 0 \Rightarrow k = -1$$

$$\therefore \text{Plane is } (2x + 3y + 4z - 7) - 1(x + y + z - 1) = 0$$

$$\Rightarrow x + 2y + 3z - 6 = 0$$

11. Equation of yz-plane is $x = 0$, Equation of xz-plane is $y = 0$

$\therefore$ Equation of the bisectors of the angles between

$$\text{the planes are } \frac{x}{1} = \pm \frac{y}{1} \Rightarrow x + y = 0, x - y = 0$$



LEVEL-III

1. If the angles made by the normal of the plane $2x + 3y - 4z - 16 = 0$ with the coordinates axes X, Y, Z are $\cos^{-1}k_1, \cos^{-1}k_2, \cos^{-1}k_3$, then k_1, k_2, k_3 respectively are

$$1) \frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}} \quad 2) \frac{2}{\sqrt{29}}, -\frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}}$$

$$3) \frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, -\frac{4}{\sqrt{29}} \quad 4) \frac{1}{2}, \frac{1}{3}, -\frac{1}{4}$$

2. The two planes represented by $12x^2 - 2y^2 - 6z^2 - 7yz + 6zx - 2xy = 0$ are

$$1) 2x + y + 2z = 0, 6x - 2y + 3z = 0$$

$$2) 2x - y + 2z = 0, 6x + 2y - 3z = 0$$

$$3) 2x - y + 2z + 4 = 0, 6x + 2y - 3z = 0$$

$$4) 2x - y + 2z = 0, 6x + 2y - 3z + 1 = 0$$

3. If the plane $4(x-1)+k(y-2)+8(z-5)=0$ contains the line $\frac{x-1}{2}=\frac{y-2}{4}=\frac{z-5}{3}$, then k is
 1) 2 2) 4 3) -8 4) 8
4. If the plane $3(x-2)+(y-2)+6(z+3)=0$ contains the line $\frac{x-2}{a}=\frac{y-2}{b}=\frac{z+3}{1}$ whose inclination with X-axis is 60° , then it satisfies the equation
 1) $6a^2+36a+37=0$ 2) $36a^2+37=0$
 3) $36a^2+37a+36=0$ 4) $a+3=0$
5. The angle between the planes represented by $2x^2-6y^2-12z^2+18yz+2zx+xy=0$ is
 1) $\cos^{-1}\left(\frac{16}{21}\right)$ 2) $\cos^{-1}\left(\frac{17}{21}\right)$
 3) $\cos^{-1}\left(\frac{19}{21}\right)$ 4) $\frac{\pi}{2}$
6. If the equation of the plane passing through the line of intersection of the planes $ax+by+cz+d=0$, $a_1x+b_1y+c_1z+d_1=0$ and perpendicular to the XY-plane is $px+qy+rz+s=0$ then $S=$
 1) dc_1-d_1c 2) dc_1+d_1c
 3) dd_1+cc_1 4) $aa_1+bb_1+cc_1$
7. If the points (1, 1, -3) and (1, 0, -3) lie on opposite sides of the plane $x+y+3z+d=0$ then
 1) $d < 7$ 2) $d > 8$
 3) $7 < d < 8$ 4) $d < 7$ or $d > 8$
8. P is a point such that the sum of the squares of its distances from the planes $x+y+z=0$, $x+y-2z=0$, $x-y=0$ is 5 then the locus of P is
 1) $x^2+y^2+z^2=10$ 2) $x^2+y^2+z^2=25$
 3) $x^2+y^2+z^2=5$ 4) $x^2+y^2+z^2=50$
9. The areas of triangles formed by a plane with the positive X, Y, Z, X axes respectively are 12, 9, 6 square units then the equation of the plane is
 1) $\frac{x}{4}+\frac{y}{6}+\frac{z}{3}=1$ 2) $\frac{x}{6}+\frac{y}{3}+\frac{z}{4}=1$
 3) $\frac{x}{4}+\frac{y}{4}+\frac{z}{6}=1$ 4) $\frac{x}{3}+\frac{y}{6}+\frac{z}{4}=1$

10. The plane $ax+by+cz+(-3)=0$ meet the co-ordinate axes in A, B, C. Then centroid of the triangle is
 1) (3a, 3b, 3c) 2) $\left(\frac{3}{a}, \frac{3}{b}, \frac{3}{c}\right)$
 3) $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$ 4) $\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right)$
11. Equation of the plane passing through the point (-1, 3, 2) and perpendicular to each of the planes $x+2y+3z=5$ and $3x+3y+z=0$ is
 1) $7x+8y-3z=0$ 2) $7x-8y-3z=-37$
 3) $7x-8y+3z+25=0$ 4) $7x+8y+3z=23$
12. If $P=(0,1,0)$ and $Q=(0,0,1)$ then the projection of PQ on the plane $x+y+z=3$ is
 1) 2 2) $\sqrt{2}$ 3) 3 4) $\sqrt{3}$
13. A parallelepiped is formed by the planes drawn through the points (2, 3, 5) and (5, 9, 7) parallel to the coordinate planes. The length of diagonal of the parallelepiped is
 1) 7 2) $\sqrt{38}$ 3) $\sqrt{155}$ 4) $\sqrt{7}$

LEVEL-III - KEY

- 1) 3 2) 2 3) 3 4) 1 5) 1 6) 1
 7) 3 8) 3 9) 1 10) 4 11) 3 12) 2
 13) 1

LEVEL-III - HINTS

1. The d.r's of normal of the plane $2x+3y-4z-16=0$ are (2, 3, -4) and the d.c's are $\left(\frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, \frac{-4}{\sqrt{29}}\right)$
 $\therefore \cos \alpha = \frac{2}{\sqrt{29}}, \cos \beta = \frac{3}{\sqrt{29}}, \cos \gamma = \frac{-4}{\sqrt{29}}$
 Where α, β, γ are the angle made by the normal with X, Y, Z axes respectively.
 $\therefore \alpha = \cos^{-1}\left(\frac{2}{\sqrt{29}}\right), \beta = \cos^{-1}\left(\frac{3}{\sqrt{29}}\right)$
 $\gamma = \cos^{-1}\left(\frac{-4}{\sqrt{29}}\right)$
 $\therefore k_1, k_2, k_3$ values are $\frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, \frac{-4}{\sqrt{29}}$ respectively

2. The product of the equations of planes for the option (2) is $(2x - y + 2z)(6x + 2y - 3z) = 0$
 $12x^2 - 2y^2 - 6z^2 + 7yz + 6xz - 2xy = 0$
 $\therefore$ correct answer is (2)
3. If a plane contains the line, then its normal and line are perpendicular
 $\therefore 4(2) + k(4) + 8(3) = 0$
(i.e., $al + bm + cn = 0$)
 $8 + 4k + 24 = 0 \Rightarrow k = -8$
4. The d.r.'s of line are $(a, b, 1)$
 $\therefore$ The d.c.'s of line are
 $\left(\frac{a}{\sqrt{a^2 + b^2 + 1}}, \frac{b}{\sqrt{a^2 + b^2 + 1}}, \frac{1}{\sqrt{a^2 + b^2 + 1}} \right)$
 $\therefore \cos 60^\circ = \frac{a}{\sqrt{a^2 + b^2 + 1}} \Rightarrow \frac{1}{2} = \frac{a}{\sqrt{a^2 + b^2 + 1}}$
i.e., $2a = \sqrt{a^2 + b^2 + 1}$
i.e., $3a^2 = b^2 + 1$ (1)
 As the plane contains the given line, we have
 $3(a) + 1(b) + 6(1) = 0$ (2)
 Eliminating b from (1) & (2),
 we get $3a^2 = \{-(6 + 3a)\}^2 + 1$
i.e., $6a^2 + 36a + 37 = 0$
5. The equation $2x^2 - 6y^2 - 12z^2 + 18yz + 2zx + xy = 0$ represents two planes and they are
 $2x - 3y + 6z = 0$ (1)
 $x + 2y - 2z = 0$ (2)
 If θ is the angle between the planes (1) and (2)
 then $\cos \theta = \frac{|(2)(1) + (-3)(2) + 6(-2)|}{\sqrt{2^2 + (-3)^2 + 6^2} \sqrt{1^2 + 2^2 + 2^2}} = \frac{16}{21}$
 $\therefore \theta = \cos^{-1} \frac{16}{21}$
6. The plane equation is $(ax + by + cz + d) - \left(\frac{c}{c_1} \right)$
 $(a_1x + b_1y + c_1z + d_1) = 0$.
7. d-7 and d-8 must have opposite signs $\Rightarrow 7 < d < 8$.

8. $\left(\frac{x_1 + y_1 + z_1}{\sqrt{3}} \right)^2 + \left(\frac{x_1 + y_1 - 2z_1}{\sqrt{6}} \right)^2 + \left(\frac{x_1 - y_1}{\sqrt{2}} \right)^2 = 5$
 $\Rightarrow x^2 + y^2 + z^2 = 5$
9. $\frac{1}{2}|ab| = 12, \quad \frac{1}{2}|bc| = 9, \quad \frac{1}{2}|ca| = 6$
10. A plane meet co-ordinate axes at
 $A\left(\frac{3}{a}, 0, 0\right), B\left(0, \frac{3}{b}, 0\right), C\left(0, 0, \frac{3}{c}\right)$
 $\therefore$ Centroid $G = \left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c} \right)$
11. Equation of the plane is
 $a(x + 1) + b(y - 3) + c(z + 2) = 0$
 $a + 2b + 3c = 0; 3a + 3b + c = 0$
12. If L, M are the feet of the perpendiculars from P, Q to the plane then projection of PQ is LM.
13. The lengths of edges are $a = 5 - 2 = 3$,
 $b = 9 - 3 = 6, c = 7 - 5 = 2$
 $\therefore$ Length of the diagonal $= \sqrt{a^2 + b^2 + c^2} = 7$



LEVEL-IV

1. Match the following
- | List-I | List-II |
|---|-------------------|
| a) The image of $(2, 1, 4)$ in the plane $2x - y + z + 5 = 0$ | i) $(-2, 3, 2)$ |
| b) The foot of perpendicular from $(2, 1, 4)$ in the plane $2x - y + z + 5 = 0$ | ii) $(-4, 1, 4)$ |
| c) The point lying on $2x - y + z + 5 = 0$ | iii) $(-6, 5, 0)$ |
- Which of the following is correct?
 1) a-iii, b-ii, c-i 2) a-i, b-iii, c-ii
 3) a-i, b-ii, c-iii 4) a-iii, b-i, c-ii
2. Observe the following Statements:
Statement I : The image of $(0, 0, 0)$ in the plane $2x - 3y + 6z - 49 = 0$ is $(4, -6, 12)$
Statement II : The foot of the perpendicular from $(1, 2, 3)$ to the plane $x + y - 2z - 9 = 0$ is $(3, 4, -1)$
 Which of the following is correct?
 1) I is true, II is true 2) I is true, II is false
 3) Both I and II are false 4) I is false, II is true

3. α, β, γ are the angles made by the normal of the plane $2x - 3y + 6z - 6 = 0$ with axes X, Y, Z respectively. Match the cosine of the angles given in List I with corresponding values given in List II.

List-I

a) $\cos \alpha$

b) $\cos \beta$

c) $\cos \gamma$

List-II

i) $6/7$

ii) $-3/7$

iii) $2/7$

The correct match from list I to list II is

1) a-i, b-ii, c-iii

2) a-ii, b-i, c-iii

3) a-ii, b-iii, c-i

4) a-iii, b-ii, c-i

4. The coordinates of a point P are (1, 2, -1). Match the plane given in List I with corresponding perpendicular distance from the point P to their planes given in List II

List-I

a) $2x + y - z + 2 = 0$

b) $x - y + 6z = 0$

c) $3x + 2y + 6z + 6 = 0$

List-II

i) 1

ii) $\frac{7}{\sqrt{6}}$

iii) $\frac{7}{\sqrt{38}}$

Which of the following is correct?

1) a-ii, b-i, c-iii

2) a-i, b-ii, c-iii

3) a-iii, b-ii, c-i

4) a-ii, b-iii, c-i

ASSERTION & REASON

The following questions consist of two statements one labelled as Assertion (A) and the other labelled as Reason (R). You are to examine these two statements carefully and decide if the Assertion (A) and the Reason (R) are individually true and if so, whether the Reason (R) is the correct explanation for the given Assertion (A). Select your answer to these items using the codes given below and then select the correct option.

Codes:

(A) Both A and R are individually true and R is the correct explanation of A

(B) Both A and R are individually true but R is not the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

5. Assertion (A) : The equation of the plane through the intersection of the planes $x + y + z = 6$ and

$$2x + 3y + 4z + 5 = 0 \text{ and the point}$$

$$(4, 4, 4) \text{ is } 29x + 23y + 17z = 276$$

Reason (R) : Equation of the plane through the line of intersection of the planes $P_1 = 0$ and $P_2 = 0$ is $P_1 + \lambda P_2 = 0$ ($\lambda \neq 0$)

1) A 2) B 3) C 4) D

6. Assertion (A) : The points (2, 1, 5) and (3, 4, 3) lie on opposite side of the plane $2x + 2y - 2z - 1 = 0$

Reason (R) : The algebraic perpendicular distance from the given points to the line have opposite sign

1) A 2) B 3) C 4) D

Linked comprehensive type passage:

Let two planes $p_1 : 2x - y + z = 2$, and

$p_2 : x + 2y - z = 3$ are given.

7. The equation of the plane passing through the intersection of P_1 and P_2 and the point (3, 2, 1) is
- 1) $3x - y + 2z - 9 = 0$ 2) $x - 3y + 2z + 1 = 0$
 3) $2x - 3y + z - 1 = 0$ 4) $4x - 3y + 2z - 8 = 0$
8. Equation of the plane which passes through the point (-1, 3, 2) and is perpendicular to each of the planes P_1 and P_2 is
- 1) $x + 3y - 5z + 2 = 0$ 2) $x + 3y + 5z - 18 = 0$
 3) $x - 3y - 5z + 20 = 0$ 4) $x - 3y + 5z = 0$
9. The equation of the acute angle bisector of planes P_1 and P_2 is
- 1) $x - 3y + 2z + 1 = 0$ 2) $3x + y - 5 = 0$
 3) $x + 3y - 2z + 1 = 0$ 4) $3x + z + 7 = 0$
10. The equation of the bisector of angle of the planes P_1 and P_2 which not containing origin is
- 1) $x - 3y + 2z + 1 = 0$ 2) $x + 3y = 5$
 3) $x + 3y + 2z + 2 = 0$ 4) $3x + y = 5$
11. The image of plane P_1 in the plane mirror P_2 is
- 1) $x + 7y - 4z + 5 = 0$ 2) $3x + 4y - 5z + 9 = 0$
 3) $7x - y + 2z - 9 = 0$ 4) $7x + y + 9z + 9 = 0$

LEVEL-IV-KEY

- 1) 4 2) 1 3) 4 4) 4 5) 1 6) 1
 7) 2 8) 3 9) 1 10) 4 11) 3

LEVEL-IV-HINTS

1. a) Let the image of (2, 1, 4) in the plane $2x - y + z + 5 = 0$ be (x_1, y_1, z_1) then

$$\frac{x_1 - 2}{2} = \frac{y_1 - 1}{-1} = \frac{z_1 - 4}{1} = \frac{-2\{2 \times 2 - 1 + 4 + 5\}}{(2)^2 + (-1)^2 + (1)^2} = -4$$

$$x_1 = -8 + 2 = -6; y_1 = 4 + 1 = 5; z_1 = -4 + 4 = 0$$

$\therefore$ The image is (-6, 5, 0)

- b) Since the image point of (2, 1, 4) is (-6, 5, 0), the mid point of them is the foot of the perpendicular

i.e., (-2, 3, 2)

- c) As the matches (-4, 1, 4), (-6, 5, 0) + 5 = 0 is (-4, 1, 4) verification is $2(-4) - 1 + 4 - 5 = 0$

2. I : Let the image of (0, 0, 0) in the plane $2x - 3y + 6z - 49 = 0$ be (x_1, y_1, z_1) , then

$$\frac{x_1 - 0}{2} = \frac{y_1 - 0}{-3} = \frac{z_1 - 0}{6} = \frac{2(0 - 0 + 0 - 49)}{(2)^2 + (-3)^2 + (6)^2} = 2$$

$$\therefore x_1 = 4, y_1 = 6, z_1 = 12$$

The image point is (4, -6, 12)

It is true

II : The point (3, 4, -1) lies on $x + y - 2z - 9 = 0$

$$(\because 3 + 4 + 2 - 9 = 0)$$

The d.r's of normal are (1, 1, -2) and that of line joining (1, 2, 3), (3, 4, -1) are (2, 2, -4). Hence II is also true.

3. The normal d.r's are (2, -3, 6)

$$\therefore \text{Its d.c's are } \left(\frac{2}{7}, \frac{-3}{7}, \frac{6}{7}\right)$$

$$\therefore \cos \alpha = \frac{2}{7}, \cos \beta = \frac{-3}{7}, \cos \gamma = \frac{6}{7}$$

$\therefore$ a-ii, b-ii, c-i

4. a) The perpendicular distance from (1, 2, -1) to $2x + y - z + 2 = 0$ is

$$\frac{|2(1) + (1)(2) - (-1) - 2|}{\sqrt{2^2 + (1)^2 + (-1)^2}} = \frac{7}{\sqrt{6}}$$

- b) Similarly for the plane $x - y + 6z = 0$

$$\text{perpendicular distance} = \frac{|1 - 2 + 6(-1)|}{\sqrt{(1)^2 + (-1)^2 + (6)^2}} = \frac{7}{\sqrt{38}}$$

- c) For the plane $3x + 2y + 6z + 6 = 0$

$$\text{perpendicular distance} = \frac{|3(1) + 2(2) + 6(-1) + 6|}{\sqrt{(3)^2 + (2)^2 + (6)^2}} = 1$$

5. Equation of the plane passes through the line of intersection of planes

$$x + y + z = 6 \text{ and } 2x + 3y + 4z + 5 = 0 \text{ is}$$

$$(x + y + z - 6) + \lambda(2x + 3y + 4z + 5) = 0.$$

$$\text{It passes through } (4, 4, 4) \Rightarrow \lambda = \frac{-6}{41}$$

6. The distances from the points to the plane, we

$$\text{get } \frac{-5}{2\sqrt{3}} \text{ and } \frac{7}{2\sqrt{3}}$$

7. $p_1 + \lambda p_2 = 0$ passes through (3, 2, 1)

$$\Rightarrow \lambda = -1$$

8. Equation of the plane passes through (-1, 3, 2)

$$\text{is } a(x + 1) + b(y - 3) + c(z - 2) = 0 \dots (1)$$

(1) is perpendicular to p_1 and p_2 we get

$$\frac{a}{-1} = \frac{b}{3} = \frac{c}{5}$$

9. $(-2)(-1) + (1)(-2) + (-1)(1) = -1 < 0$

acute angle bisector is

$$(-2x + y - z + 2) = (-x - 2y + z + 3)$$

10. Equation of bisector not containing the origin is

$$\frac{(-2x + y - z + 2)}{\sqrt{6}} = -\frac{(-x - 2y + z + 3)}{\sqrt{6}}$$

11. The image of the plane p_1 in the plane mirror p_2 then

$$2(2(1) + -1(2) + 1(-1))(x + 2y - z - 3) =$$

$$(1 + 4 + 1).(2x - y + z - 2)$$